

Professional Labor Markets And Public-Sector Design

Assignment Rules, Career Ladders, And Public Objectives

Special Topic 7: Labor Market Design, Contracting, and Mechanism Design – Week 5

Central question

- What can structured professional and public-sector labor markets teach us about mechanism design under staffing, fairness, transparency, and career constraints?
- Week 5 keeps the labor object visible: workers are assigned to jobs, training paths, units, schools, and promotion ladders.
- The design problem is not only stability or strategy-proofness.
- It is rule $\rightarrow$ labor margin $\rightarrow$ public objective $\rightarrow$ worker welfare $\rightarrow$ dynamic career response.

Professional and public-sector design principles

Institutional feature	Labor margin	Design implication
Compressed wages	nonprice allocation	assignment and priorities matter more
Public staffing targets	location/unit fill	efficiency is not only preference rank
Transparency requirements	legitimacy and fairness	rules must be explainable
Service obligations	entry, retention, exit	contracts change participation
Training pipelines	career value	initial matches are dynamic
Internal ladders	mobility, promotion	reassignment rules matter after entry

Rule vector and welfare objects

$$\rho = (A, P, C, L)$$

- A : assignment, clearing, rotation, or transfer rule.
- P : priority system, eligibility, reserves, scoring, staffing weights.
- C : contract terms, service obligations, probation, wage rigidity.
- L : internal ladder, promotion, transfer, training, on-the-job search.

$$W_i(\rho) = u_i(j_i) + \tau_i(j_i, \rho) + g_i(L, \rho) - c_i(j_i, \rho) - \lambda_i r_i(\rho) + b_i^{out}(\rho)$$

- Worker welfare includes assignment, training, mobility, risk, and outside options.

Public objective and institutional detail

$S(\rho)$ = match quality+priority-site staffing+public-service output+retention—unfairness or opacity costs

- The same matching rule can behave differently across residency, schools, civil service, and military labor markets.
- Wage rigidity makes priorities and assignment rules substitute for prices.
- Service obligations change outside options and continuation value.
- Staffing targets change what “efficient” means.
- Training and promotion systems make initial assignment dynamically consequential.

Teacher and public-service assignment

- Teacher placement is a labor-market design problem, not only a school-choice analogy.
- The rule affects worker location choices, match quality, retention, training value, and public-service output.
- Public-service settings require explicit tradeoffs between teacher preferences, school priorities, geographic staffing, transparency, and student access.
- Recruitment contracts affect who enters; assignment rules affect where they work; career ladders affect whether they stay.
- Empirical design must separate worker preference satisfaction from service delivery and staffing equity.

Internal ladders, training, and on-the-job search

Dynamic margin	Observable object	Design question
Training	credentials, completion, specialization	did assignment build human capital?
Promotion	rank, wage grade, leadership role	did the rule change career paths?
Transfer	internal moves, bids, rotations	can mismatch be repaired?
On-the-job search	applications, bids, exit threats	does the ladder retain workers?
Managerial discretion	rankings, overrides, evaluations	information or favoritism?

- Initial match quality is incomplete if later mobility, promotion, and training value change.

Army and military assignment design

- Military assignment combines preferences, priorities, contracts, branch capacity, service obligations, and career pipelines.
- Branch-of-choice contracts show why matching with contracts is a labor-market object.
- Deferred acceptance can improve preference alignment, but implementation depends on branch priorities, manager behavior, and strategic communication.
- Assignment affects training, specialization, promotion, retention, and later internal mobility.
- The welfare claim must distinguish officer welfare, unit staffing, organizational mission, and long-run career outcomes.

In and out of public employment

- Public-sector mechanisms operate only through workers who apply, accept, stay, transfer, or exit.
- Wage scales, pensions, security, mission value, probation, and promotion rules shape selection into the applicant pool.
- Service obligations and internal mobility affect whether workers remain after mismatch.
- Private-sector options affect sorting, wage pressure, and the interpretation of retention.
- Entry and exit are mechanism-design margins, not background attrition.

Methods and data

Design	Data	Identifies
Mechanism randomization	market-by-market assignments	effect of rule change
Administrative redesign	preferences, assignments, outcomes	before/after rule differences
Ranking or score cutoff	applicant scores and later outcomes	local effect of entry or priority
Recruitment randomization	applicants plus performance	selection versus effort
Internal market data	bids, transfers, promotions	career mobility and mismatch repair
Surveys plus admin data	preferences and realized moves	objective alignment

- Best papers name the rule, labor margin, counterfactual, welfare object, public objective, and main threat.

- Primary anchor: deferred acceptance in Army officer labor markets.
- Challenge anchor: matching Teach For America teachers to schools.
- Reproduce: compare deferred-acceptance-style and manager-directed assignment in synthetic structured labor-market data.
- Diagnose: separate preference rank, staffing, priority alignment, training value, retention, internal mobility, and outside options.
- Transfer: adapt the design to teacher placement, public-service recruitment, civil-service entry, internal talent markets, or public-sector exit.
- Deliverable: rule, labor margin, counterfactual, data requirement, welfare object, public objective, and main threat.

Takeaways

- Professional and public-sector labor markets make design choices visible.
- Assignment rules interact with wage rigidity, service obligations, training, internal ladders, and outside options.
- Military and teacher labor markets are empirical laboratories for staffing, fairness, transparency, and worker welfare.
- A credible welfare claim must track both public objectives and worker career outcomes.
- Week 6 turns these ingredients into full empirical research designs.